

Specifications

General			
Input Channels		One	
Input Types	Digital (S³L)	Serial ASCII, TTL level, 9600 bps	
	Frequency	Range	0.5 to 1500 Hz
		Accuracy	0.5% of reading
Measurement Types		Flow, pH/ORP, Conductivity/Resistivity, Salinity, Pressure, Temperature, Level or user-defined (via 8058)	
Enclosure and Display			
Case Material		PBT	
Window		Shatterproof glass	
Keypad		4 buttons, injection-molded silicone rubber seal	
Display		Backlit, 7 and 14-segment	
Update Rate		1 s	
LCD Contrast		5 settings	
Indicators		“Dial-type” digital bar graph. LEDs for Open Collector, Relays and Warning Indicator	
Enclosure Size		¼ DIN	
Mounting	Panel	¼ DIN, ribbed on four sides for panel mounting clip inside panel, silicon gasket included	
	Field	Mounts to standard Signet field mount junction boxes. Optional angle adjustment adapter available.	
	Wall	Large enclosure (sold as an accessory) that encases the panel mount transmitter	
Display Ranges			
pH		0.00 to 15.00 pH	
pH Temperature		-99 °C to 350 °C	-146 °F to 662 °F
ORP		-1999 to 1999.9 mV	
Flow Rate		-9999 to 99999 units per second, minute, hour or day	
Totalizer		0.00 to 99999999 units	
Conductivity		0.0000 to 99999 µS, mS, PPM and PPB (TDS), kΩ, MΩ	
Conductivity Temperature		-99 °C to 350 °C	-146 °F to 662 °F
Temperature		-99 °C to 350 °C	-146 °F to 662 °F
Pressure		-40 to 1000 psi	
Level		-9999 to 99999 m, cm, ft, in, %	
Volume		0 to 99999 cm³, m³, in³, ft³, gal, L, lb, kg, %	
Salinity		0 to 100 PPT	
Environmental			
Ambient Operating Temperature			
Backlit LCD		-10 °C to 70 °C	14 °F to 158 °F
Storage Temperature		-15 °C to 80 °C	5 °F to 176 °F
Relative Humidity		0 to 100% condensing for field mount; 0 to 95% non-condensing for panel mount	
Maximum Altitude		4,000 m (13,123 ft)	
Enclosure Rating		Designed to meet NEMA 4X/IP65 (front face only on panel mount); field mount is 100% NEMA 4X/IP65	

Specifications (continued)

Electrical Requirements

Power to Sensors

Voltage	+4.9 to 5.5 VDC @ 25 °C, regulated		
Current	1.5 mA max in loop power mode (up to 2.0 mA with 24 V @ 300 Ω max. loop impedance); 20 mA max when using DC power		
Short Circuit	Protected		
Isolation	Low voltage (< 48V AC/DC) to loop with DC power connected		
No isolation when using loop power only			
Terminal Blocks	Pluggable screw type	14 AWG max wire gauge	

Input Power

DC	12 to 32 VDC ±10%, regulated
Overvoltage Protection	48 Volt Transient Protection Device
Current limiting for circuit protection	
Reverse-Voltage Protection	

Loop Power

No DC Power Input

Max. Loop Impedance	50 Ω @ 12 V	325 Ω @ 18 V	600 Ω @ 24 V
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With DC Power Input

Max. Loop Impedance	250 Ω @ 12 V	500 Ω @ 18 V	750 Ω @ 24 V
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Relay Specifications

Dry-Contact Relays		2	Open Collector	1
	Type	SPDT	N/A	
	Form	C	N/A	
Max. Current Rating		5 A resistive	50 mA DC	
Max. Voltage Rating		30 VDC or 250 VAC		
Hysteresis		Adjustable (absolute in engineering units) (EUs)		
Latch		Reset in test screen only		
Delay		9999.9 seconds (max.)		
Test Mode		Set On or Off		
Cycle Time		99999 seconds (max.)		
Maximum Pulse Rate		400 pulses/minute		
Proportional Pulse		400 pulses/minute		
Volumetric Pulse Width		0.1 to 3200 s		
Pulse Width Modulation		0.1 to 320 s		

Input Types

Digital (S³L) or AC frequency

4 to 20 mA input via the 8058

pH/ORP input via the Digital (S³L) output from the 2750 pH/ORP Sensor Electronics

Raw Conductivity/Resistivity input directly from Signet Conductivity/Resistivity electrodes via Direct Conductivity/Resistivity Module or via 2850

Input Specifications

Digital (S ³ L)		Serial ACSII, TTL level, 9600 bps
Frequency Input		
	Sensitivity	80 mV @ 5 Hz, gradually increasing with frequency
	Span	0.5 Hz to 1500 Hz @ TTL level input
	Accuracy	± 0.5% or reading max error @ 25 °C
	Resolution	1 µS
	Repeatability	± 0.2% of reading

Specifications (continued)

Input Specifications continued

Power Supply		
	Rejection	±1 µA per volt
	Short Circuit	Protected
Update Rate		(1/frequency) + 150 ms

Output Specifications

Current Output

Current Loop Output Standard	ANSI-ISA 50.00.01 Class H		
Current Output	4 to 20 mA, isolated, fully adjustable and reversible		
Span	3.8 to 21 mA		
Zero	4.0 mA factory set; user programmable from 3.8 to 5.0 mA		
Full Scale	20.00 mA factory set; user programmable from 19.0 to 21.0 mA		
Accuracy	±32 µA max. error @ 25 °C @ 24 VDC		
Resolution	6 µA or better		
Temperature Drift	±1 µA per °C		
Power Supply Rejection	±1 µA per V		
Isolation	Low voltage (< 48 VAC/DC)		
Voltage	12 to 32 VDC ±10%		
Max. Impedance (with DC power input)	250 Ω @ 12 VDC	500 Ω @ 18 VDC	750 Ω @ 24 VDC
Max. Impedance (no DC power input)	50 Ω @ 12 VDC	325 Ω @ 18 VDC	600 Ω @ 24 VDC
Update Rate	150 mS nominal		
Short circuit and reverse polarity protected			
Adjustable Span	Reversible		
Error Condition	Selectable error condition 3.6 or 22 mA		
Actual update rate determined by sensor type			
Test Mode	Increment to desired current (range 3.8 to 21.00 mA)		
Open Collector Output	50 mA DC max., 30 VDC		

Shipping Weights

Base Unit	0.63 kg	1.38 lb
H COMM Module	0.16 kg	0.35 lb
Conductivity Module	0.16 kg	0.35 lb
Relay Module	0.19 kg	0.41 lb

Standards and Approvals

CE, UL, CUL
RoHS Compliant, China RoHS
Manufactured under ISO 9001 and ISO 14001 for Environmental Management

Dimensions

